

Task-1: Lab Setup & Environment Configuration Report

Internship: Cybersecurity & Ethical Hacking

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Task: Foundations & Environment Setup

Duration: Days 1–12

Objective

To build a secure virtual lab environment for cybersecurity practice by configuring:

- Kali Linux (Attacker Machine)
 - Metasploitable 2 (Target Machine)
 - Host-Only Network Isolation
 - Basic Tool Familiarization (Nmap, Wireshark)
-

Tools & Technologies Used

Tool	Purpose
VirtualBox	Virtualization platform
Kali Linux	Attacker system
Metasploitable 2	Vulnerable target system
Nmap	Network scanning
Wireshark	Packet capture & analysis

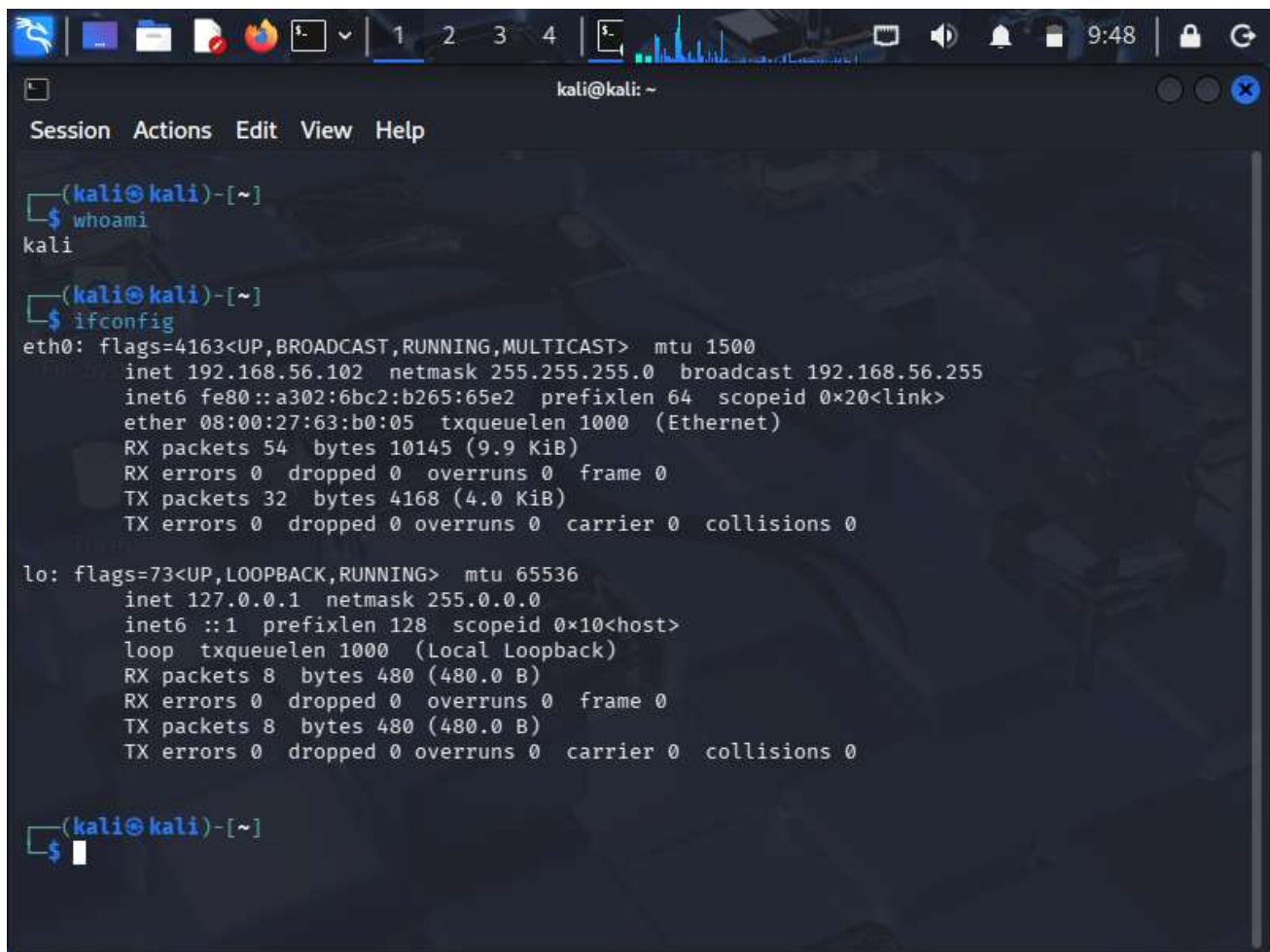
Kali Linux Setup

- Imported Kali VirtualBox image
- Allocated 4GB RAM
- Assigned 2 CPU cores
- Configured Host-Only network

Verification:

ifconfig

whoami



```
(kali@kali)-[~]
$ whoami
kali

(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.56.102  netmask 255.255.255.0  broadcast 192.168.56.255
    inet6 fe80::a302:6bc2:b265:65e2  prefixlen 64  scopeid 0x20<link>
    ether 08:00:27:63:b0:05  txqueuelen 1000  (Ethernet)
    RX packets 54  bytes 10145 (9.9 KiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 32  bytes 4168 (4.0 KiB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 8  bytes 480 (480.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 8  bytes 480 (480.0 B)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

(kali@kali)-[~]
$
```

Metasploitable Setup

- Added VM in VirtualBox
- Configured Host-Only Adapter
- Booted successfully

Verification:

Ifconfig

```

RX bytes:56737 (55.4 KB) TX bytes:56737 (55.4 KB)

msfadmin@metasploitable:~$ ifconfig
>
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:1e:e2:db
          inet addr:192.168.56.101  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe1e:e2db/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:97 errors:0 dropped:0 overruns:0 frame:0
          TX packets:63 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:13068 (12.7 KB)  TX bytes:10045 (9.8 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:170 errors:0 dropped:0 overruns:0 frame:0
          TX packets:170 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:56737 (55.4 KB)  TX bytes:56737 (55.4 KB)

msfadmin@metasploitable:~$

```

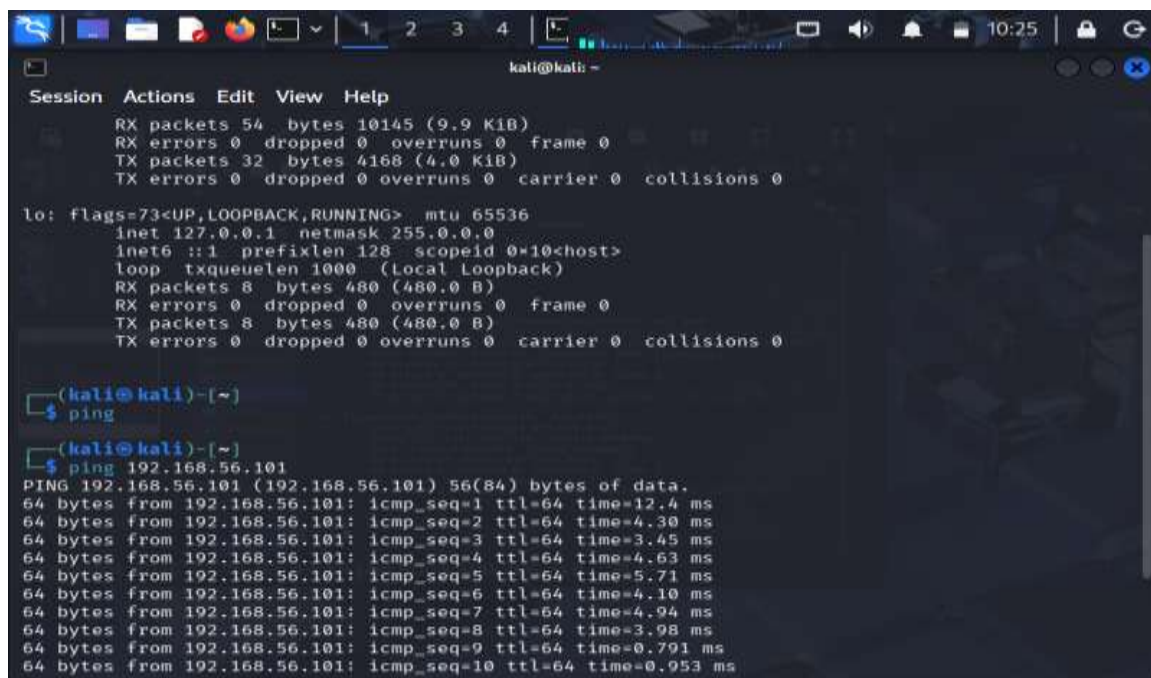
Network Verification

From Kali:

ping <metasploitable-ip>

Result:

- Successful ICMP reply
- Network connectivity confirmed



```

kali@kali: ~
Session Actions Edit View Help
RX packets 54 bytes 10145 (9.9 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 32 bytes 4168 (4.0 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)~[~]
$ ping
(kali@kali)~[~]
$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=12.4 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=4.30 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=3.45 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=4.63 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=5.71 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=4.10 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=4.94 ms
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=3.98 ms
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=0.791 ms
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=0.953 ms

```

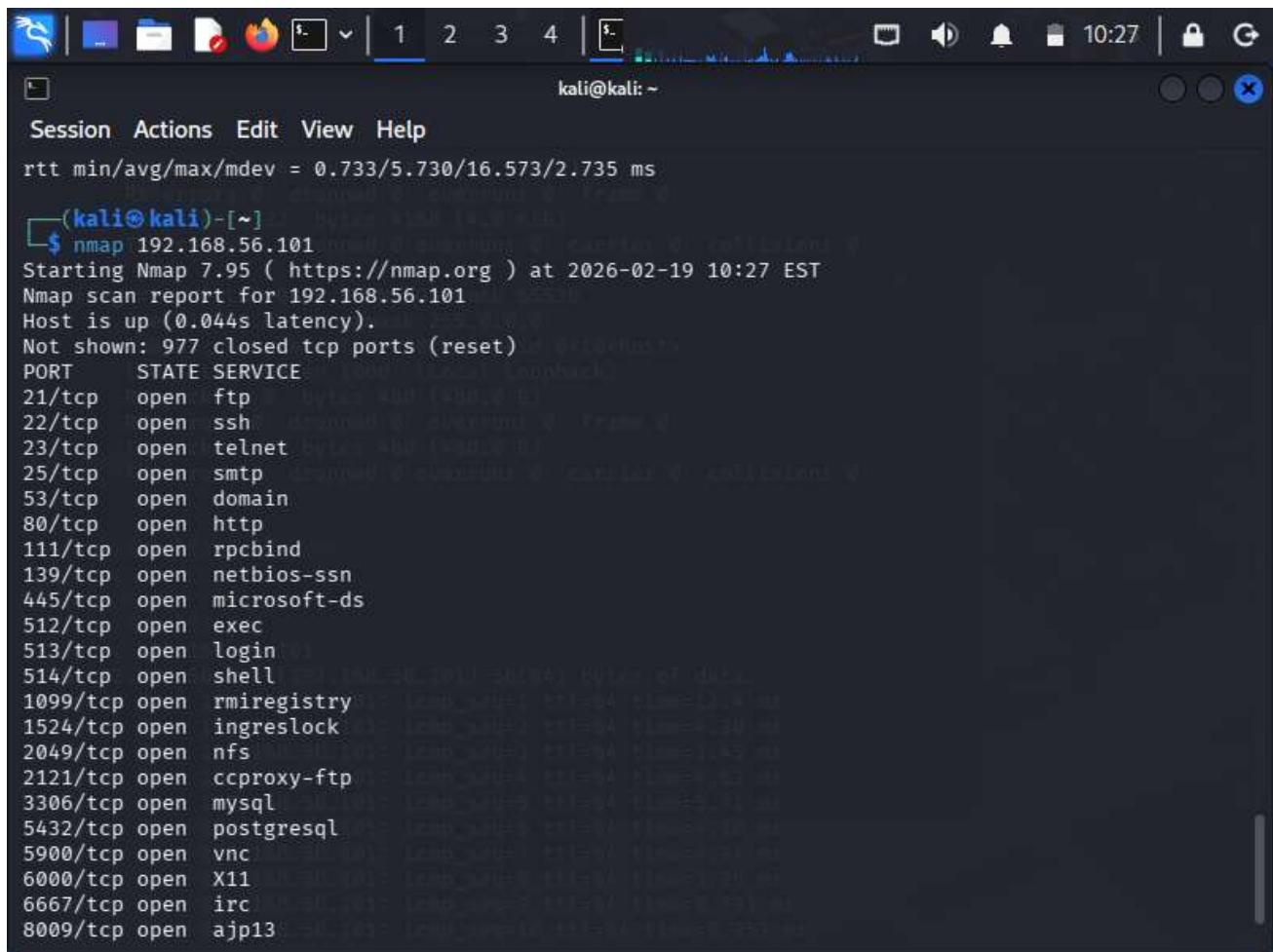
Nmap Scan Verification

Command used:

nmap <target-ip>

Observations:

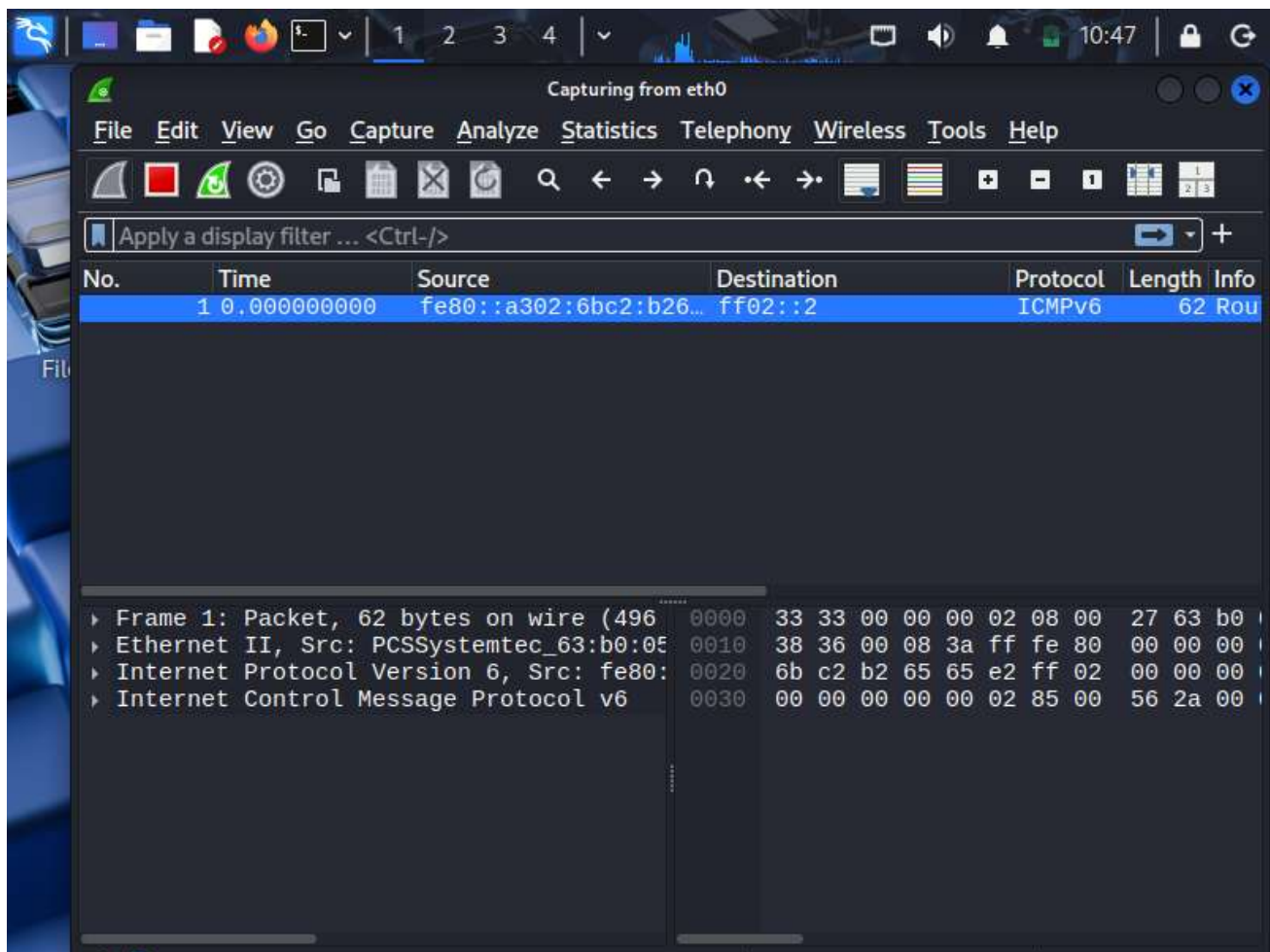
- Multiple open ports detected
- Services identified (FTP, SSH, HTTP etc.)



```
kali@kali: ~  
Session Actions Edit View Help  
rtt min/avg/max/mdev = 0.733/5.730/16.573/2.735 ms  
(kali@kali)-[~]  
$ nmap 192.168.56.101  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-19 10:27 EST  
Nmap scan report for 192.168.56.101  
Host is up (0.044s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13
```

Wireshark Packet Capture

- Started capture on Host-Only interface
- Generated ICMP traffic using ping
- Verified packet capture



Key Learnings

- Understood attacker ↔ victim lab setup
- Configured isolated private network
- Performed basic reconnaissance and scanning
- Observed live network packets
- Gained hands-on exposure to cybersecurity tools

Conclusion

The virtual cybersecurity lab was successfully configured and validated.

All required tools were tested and network communication between attacker and target systems was confirmed.

This environment will be used for further security testing tasks in upcoming internship modules.